

SREE PAVAN KOTHAPALLI

MES & Factory Automation Software Engineer | +1 (936) 666-1215 | sree.k1130@gmail.com

PROFESSIONAL SUMMARY

Software Engineer and Data Engineer with 3 years of experience developing and supporting enterprise data integration, Manufacturing Execution Systems (MES), factory automation workflows, and cloud-native applications. Experienced in MES configuration, CIM/EAP integration, manufacturing master data, equipment interfaces, quality data integration, UAT, go-live support, and production troubleshooting. Strong expertise in Python, SQL, Kafka, Azure, AWS, Databricks, and AI-enabled industrial analytics. Hands-on academic project experience in Industrial IoT, sensor data ingestion, equipment health monitoring, and predictive maintenance analytics.

TECHNICAL SKILLS

- MES & Factory Automation:** MES Configuration & Support, CIM/EAP Integration, Equipment Interfaces, Manufacturing Master Data, Traceability, Quality Data Integration, UAT, Go-Live & Post Go-Live Support
- Programming Languages:** Python, SQL, Scala, Bash, T-SQL, PowerShell
- Real-Time Integration & Automation:** Apache Kafka, Kafka Connect, Kafka Streams, Schema Registry, Azure Event Hubs, Azure Stream Analytics, Event-Driven Architecture, Enterprise Application Integration, Automated Data Processing, Real-Time Data Ingestion
- Industrial IoT & Equipment Analytics:** Azure IoT Hub, IoT Sensor Data Processing, Equipment Health Monitoring, Predictive Maintenance Analytics, Industrial Data Analytics, Power BI Monitoring Dashboards
- Data Engineering & Processing:** Apache Spark, PySpark, Databricks, Delta Lake, Spark SQL, Apache Airflow, Azure Data Factory, Hadoop, Hive, HDFS, Azure Synapse Analytics
- Cloud Platforms:** Microsoft Azure, AWS S3, AWS EMR, AWS Glue, Azure Blob Storage, Azure Data Lake Gen2, Azure Functions
- Databases:** PostgreSQL, MySQL, SQL Server, Oracle, Snowflake, Azure SQL Database
- Data Quality & Reliability:** Data Validation, Data Integrity, Data Quality Controls, Great Expectations, Error Handling, Data Consistency, Data Transformation, Performance Optimization
- DevOps & Deployment:** Git, Jenkins, Azure DevOps, Docker, Kubernetes, CI/CD Pipelines, Linux, Bash Automation
- Monitoring & Troubleshooting:** Prometheus, Grafana, Azure Monitor, System Monitoring, Alerting, Performance Troubleshooting, Operational Dashboards
- AI & Analytics Tools:** MLflow, Databricks, Power BI, Tableau, Predictive Maintenance Analytics
- Documentation:** Confluence, Draw.io, Markdown, Technical Documentation, Data Flow Diagrams

PROFESSIONAL EXPERIENCE

Data Engineer

Alephys LLC

Jun 2024 - Present

Houston, Texas

Enterprise Data Integration

- Designed and implemented scalable, real-time integration solutions using Apache Kafka, Kafka Connect, and Schema Registry for reliable, consistent data exchange across distributed applications and heterogeneous systems, handling 1M+ events per minute with sub-second latency.
- Developed automated Python/SQL workflows and real-time applications on Databricks and Delta Lake for high-volume operational data, processing 50+ TB monthly; integrated Azure Data Factory with Kafka pipelines to move data between on-premises and cloud platforms.
- Implemented validation, consistency, access-control, and encryption mechanisms to protect data integrity across distributed processing systems.
- Optimized Spark SQL queries and data models, improving performance by 50% and reducing infrastructure cost; built Jenkins/Git CI/CD pipelines for reliable, automated releases.

MES, Equipment Integration & Factory Automation

- Developed, configured, tested, and supported MES applications for automated production execution and process monitoring; translated manufacturing requirements into MES configurations, enhancements, specifications, and automated workflows.
- Integrated MES with CIM, EAP, and manufacturing data platforms for production data exchange; configured and maintained master data (process flows, equipment, recipes, materials, operational parameters) supporting work-order execution, production tracking, equipment status monitoring, and traceability.
- Developed and supported equipment interfaces for automated exchange of equipment events, process parameters, and production status; worked with equipment vendors and automation engineers to configure, test, and troubleshoot them.
- Integrated factory automation systems with enterprise data platforms for real-time equipment monitoring, analytics, and event-driven data collection that reduced manual intervention.
- Integrated manufacturing quality data with production execution for monitoring and traceability; implemented validation and reconciliation controls and supported data models for materials, recipes, process steps, equipment, and production events.

Testing, UAT, Go-Live & Collaboration

- Participated in functional and integration testing, UAT, and defect resolution; supported deployment, go-live readiness, and post-go-live stabilization.
- Investigated application failures, equipment interface errors, and data inconsistencies to minimize operational disruption; prepared test cases, deployment documentation, troubleshooting guides, and SOPs.
- Collaborated with manufacturing, quality, IT, automation, and vendor teams to deliver integrated factory software solutions, including AI-enabled analytics for equipment anomaly detection and predictive maintenance.
- Supported implementation of scalable factory systems in a greenfield manufacturing environment.

Data Engineer (Intern)

Jan 2024 – May 2024

Alephys LLC

Houston, Texas

- Built ETL pipelines (Spark, Hive) ingesting 500GB+ daily from diverse sources; partitioned Hive tables cut query times from minutes to seconds.
- Migrated 50+ tables from on-premises Hadoop to Azure Synapse with complex transformations while preserving data integrity.
- Configured Kafka producers/consumers for real-time streaming between on-premises and cloud systems with robust error handling.
- Developed Power BI dashboards connecting to Databricks and Hadoop data sources for business reporting.

Data Engineer

May 2021 – May 2022

Birlasoft

Chennai, India

- Developed automated data processing and operational workflows using Python, SQL, Linux, and Bash scripting to improve system reliability and reduce manual intervention.
- Built real-time data processing frameworks using Azure Stream Analytics and Azure Event Hubs to enable continuous ingestion and processing of event-driven data.
- Designed and implemented scalable Azure SQL data pipelines supporting enterprise operational reporting and analytics, processing 10TB+ monthly with 99.5% reliability.
- Automated data validation, backup, and system maintenance activities using Linux shell scripting.
- Configured monitoring and alerting mechanisms to identify processing failures and support timely troubleshooting of operational issues.
- Administered Linux (Oracle Linux) environments supporting mission-critical data processing applications, including performance tuning and security hardening.
- Managed Hadoop clusters for distributed processing of large financial datasets.
- Developed Power BI dashboards to visualize operational data and provide actionable insights to business stakeholders.
- Designed and optimized analytical data models in Azure Synapse, improving reporting performance by 45%.

SELECTED PROJECTS

Azure-Based Industrial IoT and Equipment Monitoring Pipeline — Master's Project

- Designed and developed an end-to-end Industrial IoT data processing solution using Azure IoT Hub, Azure Stream Analytics, and Azure Data Lake to ingest and process real-time sensor data from connected devices.
- Implemented event-driven data processing workflows to collect, transform, and store sensor-generated information for operational analytics.
- Developed Azure Functions to perform data cleansing and transformation before persisting processed sensor data in Azure Blob Storage.
- Integrated Power BI dashboards to visualize device health metrics and provide insights supporting predictive maintenance in industrial applications.
- Applied cloud-native, serverless architecture principles to develop scalable and reliable IoT data processing workflows.
- Demonstrated real-time monitoring and analytics capabilities relevant to industrial equipment performance and automated manufacturing environments.

Kafka-Driven Real-Time Data Pipeline

- Deployed a fault-tolerant, high-throughput Kafka pipeline on Cloudera with Kafka Connect source integration and KSQL DB real-time analytics.

On-Premises to Azure Data Migration

- Migrated large datasets from SQL Server to Azure Data Lake Gen2 with Databricks ETL and Synapse analytics, ensuring consistency and integrity; added Kafka streaming to cut latency.

EDUCATION

Master of Science in Computer Science, Univ. of North Texas, Denton, TX

Aug 2022 – May 2024